Introduction to R

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2026-10-17

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Preface

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Structure of the book

Chapter Introduction to R gives a quick overview of R and its evolution over the last 2 decades. Chapter Install R & RStudio acts as a guide for installing R and RStudio. Chapters Variables in R and Data Types in R introduce the reader to variables and data types in R. Chapter Getting Help in R shows the right place and right way to ask for help. Chapters Vectors in R to Lists in R introduce the basic data structures in R. Chapter Install & Update R Packages explores the R package system.

Software information

The R session information when compiling this book is shown below:

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.5 LTS
```

```
Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;
LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
 [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
 [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
[10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C

time zone: UTC
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.5.3  fastmap_1.2.0   cli_3.6.6       htmltools_0.5.9
 [5] tools_4.5.3     otel_0.2.0      yaml_2.3.12     rmarkdown_2.32
 [9] knitr_1.52      jsonlite_2.0.0  xfun_0.62       digest_0.6.39
[13] rlang_1.3.0     renv_1.2.4      evaluate_1.0.5
```

We do not add prompts (> and +) to R source code in this book, and we comment out the text output with two hashes ## by default, as you can see from the R session information above. This is for your convenience when you want to copy and run the code (the text output will be ignored since it is commented out). Package names are in bold text (e.g., **rmarkdown**), and function names are followed by parentheses (e.g., `mean()`). The double-colon operator `::` means accessing an object from a package.

About the Author

Aravind Hebbali is the founder of Rsquared Academy. He earned his Masters in Economics from Madras School of Economics. As an active R user, he has authored several R packages such as

- olsrr
- rfm
- descriptr
- blorr
- xplorerr

In 2015, he founded Rsquared Academy, a free and open source education initiative with focus on data science and analytics. Apart from self paced online courses, Rsquared Academy offers customized learning modules for corporates and universities.

You can find him on GitHub.

Introduction to R

Learning Objectives: After this chapter you will be able to:

- Describe what R is and why it is widely used for data science.
- Name the organizations that steward R and its ecosystem.
- Distinguish RStudio Desktop from Positron as an IDE for R.
- Locate the panes you will work in: script editor, console, and output.

Overview

R is a free and open source programming language widely used in academia, industry and government organisations for data science and machine learning. R is well suited for interactive data analysis, visualization and modeling. To view a list of organizations using R, click here.

Before we talk about R, let us learn a bit about the S language, which was the predecessor to R. It was developed by Richard Becker, John Chambers and Allan Wilks at the Bell Laboratories. R is a modern implementation of S.

R itself was written by Ross Ihaka and Robert Gentleman at the University of Auckland. The development of R was started in 1992 and an initial version was released in 1995. Version 1.0 was released in February 2000. Version 2.0 was released in October 2004. R is under active development by the R Core Team, with regular annual major/minor releases (such as R 4.4 and R 4.5).

R is released under GNU GPL license and is currently one of the most popular tools for data science and machine learning. You can find more details and information related to R at the official website of the R project. You can download R, learn more about the R foundation, its members, activities, get the latest news about R and also donate to the R foundation.

Organizations

R Foundation

R foundation is a not for profit organization founded by the members of the R core team and is responsible for the continued development of R. It also supports the annual useR conferences. You can learn more about the R foundation here.

R Consortium

R consortium provides support to the R foundation and other key organizations and groups developing, maintaining and using R. It funds projects that improve the infrastructure around R and provides support to conferences and user groups around the world. You can learn more about R consortium here.

Installation

R is available for Windows, Mac and is part of many Linux distributions. You can download and install R from any of the CRAN mirrors listed here.

CRAN

CRAN stands for **C**omprehensive **R** **A**rchive **N**etwork and you can download R and user contributed packages from here for different operating systems. CRAN is pronounced as KRAN or C-RAN.

Extensions

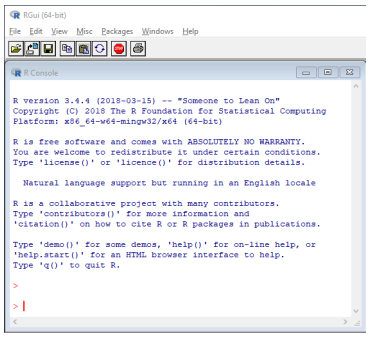
User contributed packages can be downloaded from

- CRAN
- Bioconductor
- GitHub
- R-Forge

GUI

R GUI

R comes with a simple GUI with sufficient features for data analysis and visualization.



RStudio

RStudio is a very popular IDE (integrated development environment) built to suit the data science work flow. RStudio is developed by Posit (formerly RStudio, rebranded in 2022); Positron is a modern alternative IDE for R and Python.

Install R & RStudio

Learning Objectives: After this chapter you will be able to:

- Install R on Windows, macOS, or Linux from a CRAN mirror.
- Verify an installation by running `R.version.string`.
- Install RStudio Desktop, after R itself is in place.
- Create an RStudio Project, so working directory, history, and settings travel together in a single `.Rproj` file.

R

Follow these steps:

- Download R for your operating system from a CRAN mirror (Windows, macOS, or Linux installers are listed there).
- Run the installer with default options.
- Verify the installation by opening a terminal and running `R.version.string`.

RStudio

Follow these steps:

- Download RStudio Desktop from posit.co/download/rstudio-desktop.
- Install it after installing R (RStudio needs R to work).
- Open RStudio and create a new project via *File* > *New Project* (this creates an `.Rproj` file that keeps your working directory, history, and settings together).
- Verify your setup by running `R.version.string` in the R console.

Variables in R

Learning Objectives: After this chapter you will be able to:

- Explain what a variable is and why programs use them.
- Create and reassign variables with `<-`.
- Use variables in arithmetic and in function calls.
- Name variables legally, including the rule that permits a leading `..`
- Choose `snake_case` over periods, which are legal but discouraged.

Introduction

In the previous chapter, we learnt to install RStudio. In this chapter, we will learn about variables and data types. You can skip this chapter, if you have prior experience in any other programming language.

What is a variable?

- variables are the fundamental elements of any programming language
- they are used to represent values that are likely to change
- they reference memory locations that store information/data

Let us use a simple case study to understand variables. Suppose you are computing the area of a circle whose radius is 3. In R, you can do this straight away as shown below:

```
3.14 * 3 * 3
```

```
[1] 28.26
```

But you cannot reuse the radius or the area computed in any other analysis or computation. Let us see how variables can change the above scenario and help us in reusing values and computations.

Creating Variables

A variable consists of 3 components:

- variable name
- assignment operator
- variable value

We can store the value of the radius by creating a variable and assigning it the value. In this case, we create a variable called `radius` and assign it the value `3` using the assignment operator `<-`.

```
radius <- 3
radius
```

```
[1] 3
```

Now that we have learnt to create variables, let us see how we can use them for other computations. For our case study, we will use the `radius` variable to compute the area of a circle.

Using Variables

We will create two variables, `radius` and `pi`, and use them to compute the area of a circle and store it in another variable `area`.

```
# assign value 3 to variable radius
radius <- 3

# assign value 3.14 to variable pi
pi <- 3.14

# compute area of circle
area <- pi * radius * radius

# call radius
radius
```

```
[1] 3
```

```
# call area
area
```

```
[1] 28.26
```

Components of a Variable

Naming Conventions

- Name must begin with a letter or a dot (`.`); e.g. `.hidden <- 42` is valid in base R. Do not use numbers, dollar sign (`$`) or underscore (`_`) as the first character.
- The name can contain numbers or underscore. Do not use dash (`-`). Periods (`.`) are syntactically valid in R variable names, but are discouraged in modern style guides (tidyverse style prefers `snake_case`).
- Do not use the names of keywords and avoid using the names of built in functions.

- Variables are case sensitive; `average` and `Average` would be different variables.
- Use names that are descriptive. Generally, variable names should be nouns.
- If the name is made of more than one word, use underscore to separate the words.

Data Types in R

Learning Objectives: After this chapter you will be able to:

- Create numeric, integer, character, and logical values.
- Check a value's type with `is.numeric()`, `is.integer()`, `is.character()`, and `is.logical()`.
- Convert between types with `as.integer()`, `as.character()`, and friends.
- Explain implicit coercion when a vector mixes types.
- Create and format a date, and recognise `POSIXct` for date-times.

Every value in R has a type, and the type decides what you can do with it. This chapter walks through the five you will meet first: `numeric` (double), `integer`, `character`, `logical`, and `date`/`time`. Most of the time you will let R infer the type for you, and then use the `is.*()` and `as.*()` functions to check or change it.

Numeric

In R, numbers are represented by the data type `numeric`. We will first create a variable and assign it a value. Next we will learn a few methods of checking the type of the variable.

```
# create two variables
number1 <- 3.5
number2 <- 3

# check data type
class(number1)
```

```
[1] "numeric"
```

```
class(number2)
```

```
[1] "numeric"
```

```
# check if data type is numeric
is.numeric(number1)
```

```
[1] TRUE
```

```
is.numeric(number2)
```

```
[1] TRUE
```

If you carefully observe, `integers` are also treated as `numeric/double`. We will learn to create integers in a while. In the meanwhile, we have introduced two new functions in the above example:

- `class()`: returns the `class` or `type`
- `is.numeric()`: tests whether the variable is of type `numeric`

Integer

Unless specified otherwise, integers are treated as `numeric` or `double`. In this section, we will learn to create variables of the type `integer` and to convert other data types to `integer`.

- create a variable `number1` and assign it the value `3`
- check the data type of `number1` using `class`
- create a second variable `number2` using `as.integer` and assign it the value `3`
- check the data type of `number2` using `class`
- finally use `is.integer` to check the data type of both `number1` and `number2`

```
# create a variable and assign it an integer value
number1 <- 3

# create another variable using as.integer
number2 <- as.integer(3)

# check the data type
class(number1)
```

```
[1] "numeric"
```

```
class(number2)
```

```
[1] "integer"
```

```
# use is.integer to check data type
is.integer(number1)
```

```
[1] FALSE
```

```
is.integer(number2)
```

```
[1] TRUE
```

Character

Letters, words and group of words are represented by the data type `character`. All data of type `character` must be enclosed in single or double quotation marks. In fact any value enclosed in quotes will be treated as `character`. Let us create two variables to store the first and last name of some random guy.

```
# first name
first_name <- "jovial"

# last name
```

```
last_name <- 'mann'

# check data type
class(first_name)
```

```
[1] "character"
```

```
class(last_name)
```

```
[1] "character"
```

```
# use is.character to check data type
is.character(first_name)
```

```
[1] TRUE
```

```
is.character(last_name)
```

```
[1] TRUE
```

You can coerce any data type to `character` using `as.character()`.

```
# create variable of different data types
age <- as.integer(30) # integer
score <- 9.8          # numeric/double
opt_course <- TRUE    # logical
today <- Sys.time()   # date time

as.character(age)
```

```
[1] "30"
```

```
as.character(score)
```

```
[1] "9.8"
```

```
as.character(opt_course)
```

```
[1] "TRUE"
```

```
as.character(today)
```

```
[1] "2026-10-17 05:15:52.338083"
```

Logical

Logical data types take only 2 values. Either `TRUE` or `FALSE` (plus `NA` for missing values, giving three-valued logic in R). Such data types are created when we compare two objects in R using comparison or logical operators.

- create two variables `x` and `y`
- assign them the values `TRUE` and `FALSE` respectively
- use `is.logical` to check data type
- use `as.logical` to coerce other data types to `logical`

```r
# create variables x and y
x <- TRUE
y <- FALSE

# check data type
class(x)
```

```
[1] "logical"
```

```r
is.logical(y)
```

```
[1] TRUE
```

The outcome of comparison operators is always `logical`. In the below example, we compare two numbers to see the outcome.

```r
# create two numeric variables
x <- 3
y <- 4

# compare x and y
x > y
```

```
[1] FALSE
```

```r
x < y
```

```
[1] TRUE
```

```r
# store the result
z <- x > y
class(z)
```

```
[1] "logical"
```

`TRUE` is represented by all numbers except `0`. `FALSE` is represented only by `0` and no other numbers.

```r
# TRUE and FALSE are represented by 1 and 0
as.logical(1)
```

```
[1] TRUE
```

```r
as.logical(0)
```

```
[1] FALSE
```

```r
# using numbers
as.numeric(TRUE)
```

```
[1] 1
```

```r
as.numeric(FALSE)
```

```
[1] 0
```

```r
# using different numbers
as.logical(c(-2, -1.5, -1, 0, 1, 2))
```

```
[1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
```

Use `as.logical()` to coerce other data types to `logical`.

```
# create variable of different data types
age <- as.integer(30) # integer
score <- 9.8          # numeric/double
opt_course <- TRUE    # logical
today <- Sys.time()   # date time

as.logical(age)
```

```
[1] TRUE
```

```
as.logical(score)
```

```
[1] TRUE
```

```
as.logical(opt_course)
```

```
[1] TRUE
```

```
as.logical(today)
```

```
[1] TRUE
```

Date

R has dedicated types for dates and date-times. Use `Date` for calendar dates and `POSIXct` for date-times:

```
# today's date
today_date <- Sys.Date()
today_date
```

```
[1] "2026-10-17"
```

```
class(today_date)
```

```
[1] "Date"
```

```
# create a date from a string
independence_day <- as.Date("2026-01-15")
independence_day
```

```
[1] "2026-01-15"
```

```
# pretty-print a date
format(independence_day, "%B %d, %Y")
```

```
[1] "January 15, 2026"
```

For date-times, use `as.POSIXct()`. Once you need to do arithmetic with dates — finding differences, adding months, handling time zones — reach for the lubridate package, which provides a consistent, readable vocabulary for all of it.

Getting Help in R

Learning Objectives: After this chapter you will be able to:

- Look up any known function with `?name` or `help()`.
- Search help topics when you do not know the function name.
- Read a package's documentation, examples, and vignettes.
- Decide when the built-in help system is enough and when to go online.

Introduction

In this chapter, we will learn about the different methods of getting help in R. Often we get stuck while doing some analysis as either we do not know the correct function to use or its syntax. It is important for anyone who is new to R to know the right place to look for help. There are two ways to look for help in R:

- built in help system
- online

In the first section, we will look at various ways to access the built in help system of R. In the second section, we will look at various online resources that can supplement the built in help system. Let us get started!

Online Resources

R Bloggers

R Bloggers aggregates blogs written in english from across the globe. This is the first place you want to visit if you want help with R, data analysis, visualization and machine learning. There are blogs on a wide range of topics and the latest content is delivered to your inbox if you subscribe. If you are a blogger yourself, share it with the R community by adding your blog to R Bloggers.

Stack Overflow

Stack Overflow is a great place to visit if you are having trouble with R code or packages. Chances are high that someone has already encountered the same or similar problem and you can use the answers given by R experts. In case you have encountered a new problem or issue, you can ask for help by providing a reproducible example of your analysis along with the R code. Use the reprex package to create reproducible examples.

Twitter

The R community is very active on Twitter/X and there are lot of R experts who are willing to help those who are new to R. Use the hashtag #rstats if you are asking for help or guidance on Twitter/X. Many community members are also active on Mastodon, Bluesky, and LinkedIn.

RStudio Community

RStudio Community is similar to Stack Overflow. You can ask questions related to RStudio, Shiny and tidyverse.

RStudio

RStudio has very good resources including cheatsheets, webinars and blogs.

Reddit

Reddit is another place where you can look for help. The discussions are moderated by R experts. There are subreddits for Rstats, Rlanguage, Rstudio and Rshiny.

R Weekly

Visit RWeekly to get regular updates about the R community. You can find information about new packages, blogs, conferences, workshops, tutorials and R jobs.

R User Groups

There are several R User Groups active across the globe. You can find the list here. Join the local user group to meet, discuss and learn from other R enthusiasts and experts.

Data Helpers

Data Helpers is a list of data analysts, scientists and engineers willing to offer guidance put together by Angela Bassa. Visit the website to learn more about how you can approach for help and guidance. (Note: this site appears unmaintained as of 2026; treat links there with caution.)

Internal

In this section, we will look at the following functions:

- `help.start()`
- `help()`
- `?`
- `??`
- `help.search()`
- `demo()`
- `example()`
- `library()`
- `vignette()`
- `browseVignettes()`

help.start

The `help.start()` function opens the documentation page in your browser. Here you can find manuals, reference and other materials.

```
help.start()
```

help

Use `help()` to access the documentation of functions and data sets. `?` is a shortcut for `help()` and returns the same information.

```
help(plot)
?plot
```

help.search

`help.search()` will search all sources of documentation and return those that match the search string. `??` is a shortcut for `help.search()` and returns the same information.

```
help.search('regression')
??regression
```

demo

demo displays an interactive demonstration of certain topics provided in a R package. Typing `demo()` in the console will list the demos available in all the R packages installed.

```
demo()
demo(scoping)
```

example

`example()` displays examples of the specified topic if available.

```
example('mean')
```

```
mean> x <- c(0:10, 50)

mean> xm <- mean(x)

mean> c(xm, mean(x, trim = 0.10))
[1] 8.75 5.50
```

Package Documentation

library

Access the documentation of a package using `help()` inside `library()`. The package need not be installed for accessing the documentation.

```
library(help = 'ggplot2')
```

vignette

A vignette is a long form guide to a R package. You can access the vignettes available using `vignette()`. It will display a list of vignettes available in installed packages.

```
vignette()
```

To access a specific vignette from a package, specify the name of the vignette and the package.

```
vignette('dplyr', package = 'dplyr')
```

browseVignettes

`browseVignettes()` is another way to access the vignettes in installed packages. It will list the vignettes in each package along with links to the web page and R code.

```
browseVignettes()
```

RsiteSearch

`RsiteSearch()` will search for specified topics in help pages, vignettes and task views using the search engine at this link and return the result in a browser.

```
RSiteSearch('glm')
```

Vectors in R

Learning Objectives: After this chapter you will be able to:

- Create atomic vectors of type numeric, integer, character, and logical.
- Name vector elements, and select elements by name.
- Apply arithmetic across a whole vector at once.
- Explain the coercion order `logical` → `integer` → `double` → `character`.
- Subset using positive, negative, and logical indices.
- Find missing values with `is.na()` and exclude them with `na.rm = TRUE`.

Introduction

Vector is the most basic data structure in R. It is a sequence of elements of the same data type. If the elements are of different data types, they will be coerced to a common type that can accommodate all the elements. Vectors are generally created using `c()` (concatenate function), although depending on the data type of vector being created, other methods can be used.

Numeric Vector

We will create a numeric vector using `c()` but you can use any function that creates a sequence of numbers. After that we will use `is.vector()` to check if it is a vector and `class` to check the data type.

```
# create a numeric vector
num_vect <- c(1, 2, 3)

# display the vector
num_vect
```

```
[1] 1 2 3
```

```
# check if it is a vector
is.vector(num_vect)
```

```
[1] TRUE
```

```
# check data type
class(num_vect)
```

```
[1] "numeric"
```

Let us look at other ways to create a sequence of numbers. We leave it as an exercise to the reader to understand the functions using `help`.

```
# using colon
vect1 <- 1:10
vect1
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
# using rep
vect2 <- rep(1, 5)
vect2
```

```
[1] 1 1 1 1 1
```

```
# using seq
vect3 <- seq(10)
vect3
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

Integer Vector

Creating an integer vector is similar to numeric vector except that we need to instruct R to treat the data as `integer` and not `numeric` or `double`. We will use the same methods we used for creating numeric vectors. To specify that the data is of type `integer`, we suffix the number with `L`.

```
# integer vector
int_vect <- c(1L, 2L, 3L)
int_vect
```

```
[1] 1 2 3
```

```
# check data type
class(int_vect)
```

```
[1] "integer"
```

```
# using colon
vect1 <- 1L:10L
vect1
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
# using rep
vect2 <- rep(1L, 5)
vect2
```

```
[1] 1 1 1 1 1
```

```
# using seq
vect3 <- seq(10L)
vect3
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

Character Vector

A character vector may contain a single character, a word or a group of words. The elements must be enclosed in single or double quotations.

```
# character vector
greetings <- c("hello", "good morning")
greetings
```

```
[1] "hello"        "good morning"
```

```
# check data type
class(greetings)
```

```
[1] "character"
```

Logical Vector

A vector of logical values will either contain `TRUE` or `FALSE` or both.

```
# logical vector
vect_logic <- c(TRUE, FALSE, TRUE, TRUE, FALSE)
vect_logic
```

```
[1]  TRUE FALSE  TRUE  TRUE FALSE
```

```
# check data type
class(vect_logic)
```

```
[1] "logical"
```

In fact, you can create an `integer` vector and coerce it to type `logical`.

```
# integer vector
int_vect <- rep(0L:1L, 3)
int_vect
```

```
[1] 0 1 0 1 0 1
```

```
# coerce to logical vector
log_vect <- as.logical(int_vect)
log_vect
```

```
[1] FALSE  TRUE FALSE  TRUE FALSE  TRUE
```

```
# check data type
class(log_vect)
```

```
[1] "logical"
```

Naming Vector Elements

It is possible to name the different elements of a vector. The advantage of naming vector elements is that we can later on use these names to access the elements. Use `names()` to specify the names of a vector. You can specify the names while creating the vector or add them later.

Method 1: Create vector and add names later

```
# create vector and add names later
vect1 <- c(1, 2, 3)

# name the elements of the vector
names(vect1) <- c("One", "Two", "Three")

# call vect1
vect1
##   One   Two Three
##     1     2     3
```

Method 2: Specify names while creating vector

```
# specify names while creating vector
vect2 <- c(John = 1, Jack = 2, Jill = 3, Jovial = 4)

# call vect2
vect2
##   John   Jack   Jill Jovial
##      1      2      3      4
```

Vector Coercion

Vectors are homogeneous i.e. all the elements of the vector must be of the same type. If we try to create a vector by combining different data types, the elements will be coerced to the most flexible type. The below table shows the order in which coercion occurs.

`character` data type is the most flexible while `logical` data type is the least flexible. If you try to combine any other data type with `character`, all the elements will be coerced to type `character`. In the absence of `character` data, all elements will be coerced to `numeric`. Finally, if the data does not include `character` or `numeric` types, all the elements will be coerced to `integer` type.

Case 1: Different Data Types

```
# vector of different data types
vect1 <- c(1, 1L, 'one', TRUE)

# call vect1
vect1
## [1] "1"    "1"    "one"  "TRUE"

# check data type
class(vect1)
## [1] "character"
```

Case 2: Numeric, Integer and Logical

```
# vector of different data types
vect1 <- c(1, 1L, TRUE)

# call vect1
vect1
## [1] 1 1 1

# check data type
class(vect1)
## [1] "numeric"
```

Case 3: Integer and Logical

```
# vector of different data types
vect1 <- c(1L, TRUE)

# call vect1
vect1
## [1] 1 1

# check data type
class(vect1)
## [1] "integer"
```

To summarize, below is the order in which coercion takes place:

`logical` ⟶ `integer` ⟶ `double/numeric` ⟶ `character`

R coerces toward the **most flexible** type. If a vector contains a `character` element, everything becomes `character`. Without `character`, elements coerce to `numeric`. Without `numeric`, elements coerce to `integer`.

Vector Operations

In this section, we look at simple operations that can be performed on vectors in R. Remember that the nature of the operations depends upon the type of data. Below are a few examples:

Case 1: Vectors of same length

```
# create two vectors
vect1 <- c(1, 3, 8, 4)
vect2 <- c(2, 7, 1, 9)

# addition
vect1 + vect2
## [1]  3 10  9 13

# subtraction
vect1 - vect2
## [1] -1 -4  7 -5
```

```
# multiplication
vect1 * vect2
## [1]  2 21  8 36

# division
vect1 / vect2
## [1] 0.5000000 0.4285714 8.0000000 0.4444444
```

Case 2: Vectors of different length

In the previous case, the length i.e. the number of elements in the vectors were same. What happens if the length of the vectors are unequal? In such cases, the shorter vector is recycled to match the length of the longer vector. The below example should clear this concept:

```
# create two vectors
vect1 <- c(2, 7)
vect2 <- c(1, 8, 5, 2)

# addition
vect1 + vect2
## [1]  3 15  7  9

# subtraction
vect1 - vect2
## [1]  1 -1 -3  5

# multiplication
vect1 * vect2
## [1]  2 56 10 14

# division
vect1 / vect2
## [1] 2.000 0.875 0.400 3.500
```

Missing Data

Missing data is a reality. No matter how careful you are in collecting data for your analysis, chances are always high that you end up with some missing data. In R missing values are represented by `NA`. In this section, we will focus on the following:

- test for missing data
- remove missing data
- exclude missing data from analysis

Detect missing data

We first create a vector with missing values. After that, we will use `is.na()` to test whether the data contains missing values. `is.na()` returns a logical vector equal to the length of the vector being tested. Another function that can be used for detecting missing values is `complete.cases()`. Below is an example:

```r
# vector with missing values
vect1 <- c(1, 3, NA, 5, 2)

# use is.na
is.na(vect1)
## [1] FALSE FALSE  TRUE FALSE FALSE

# use complete.cases
complete.cases(vect1)
## [1]  TRUE  TRUE FALSE  TRUE  TRUE
```

Omit missing data

In the presence of missing data, all computations in R will return `NA`. To avoid this, we might want to remove the missing data before doing any computation. `na.omit()` will remove all missing values from the data. Let us look at an example:

```r
# vector with missing values
vect1 <- c(1, 3, NA, 5, 2)

# call vect1
vect1
## [1]  1  3 NA  5  2

# omit missing values
na.omit(vect1)
## [1] 1 3 5 2
## attr(,"na.action")
## [1] 3
## attr(,"class")
## [1] "omit"
```

Exclude missing data

To exclude missing values from computations, use `na.rm` and set it to `TRUE`.

```r
# vector with missing values
vect1 <- c(1, 3, NA, 5, 2)

# compute mean
mean(vect1)
## [1] NA

# compute mean by excluding missing value
mean(vect1, na.rm = TRUE)
## [1] 2.75
```

> **Common Mistake:** `mean()` on a vector containing `NA` returns `NA`, not the mean of the values that are present. Add `na.rm = TRUE` to drop missing values first, and check whether dropping them is defensible for your data.

Index Vectors

One of the most important steps in data analysis is selecting a subset of data from a bigger data set. Indexing helps in retrieving values individually or a set of values that meet a specific criteria. In this chapter, we look at various ways of indexing/subsetting vectors.

`[]` is the index operator in R. We can use various expressions within `[]` to subset data. In R, **index positions begin at 1 and not 0**. To begin with, let us look at values in different index positions:

```
# fixed seed so this chapter re-renders deterministically
set.seed(42)

# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  1  5 10  8  2  4  6  9  7  3
```

```
# return third element
vect1[3]
```

```
[1] 10
```

```
# return seventh element
vect1[7]
```

```
[1] 6
```

Out of range index

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  8  7  4  1  5 10  2  6  9  3
```

```
# return value at index 0
vect1[0]
```

```
integer(0)
```

```
# length of the vector
length(vect1)
```

```
[1] 10
```

```
# out of range index
vect1[11]
```

```
[1] NA
```

In the first case, we specified the index as 0 and in the second case we used the index 11, which is greater than the length of the vector. R returns an empty vector in the first case and `NA` in the second case.

Negative Index

Using a negative index will delete the value in the said index position. Unlike other languages, it will not index elements from the end of the vector counting backwards. Let us look at an example to understand how negative index works in R:

> **Common Mistake:** `vect1[-3]` does not mean "the third element from the end". A negative index **deletes** that position. Use `tail(vect1, 3)` when you actually want the last three elements.

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  9 10  3  4  5  6  1  2  8  7
```

```
# drop third element
vect1[-3]
```

```
[1]  9 10  4  5  6  1  2  8  7
```

```
# drop seventh element
vect1[-7]
```

```
[1]  9 10  3  4  5  6  2  8  7
```

Subset Multiple Elements

If we do not specify anything within `[]`, all the elements in the vector will be returned. We can specify the index elements using any expression that generates a sequence of integers. Let us look at a few examples:

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  3  1  2  6 10  8  4  5  7  9
```

```
# return all elements
vect1[]
```

```
 [1]  3  1  2  6 10  8  4  5  7  9
```

```
# return first 5 values
vect1[1:5]
```

```
[1]  3  1  2  6 10
```

```
# return all values from the 5th position
end <- length(vect1)
vect1[5:end]
```

```
[1] 10  8  4  5  7  9
```

If you are using the colon to generate the index positions, you will have to specify both the starting and ending position, else, R will return an error.

What if we want elements that are not in a sequence as we saw in the last example? In such cases, we have to create a vector using `c()` and use it to extract elements from the original vector. Below is an example:

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  5  4  2  8  3  1 10  7  6  9
```

```
# extract 2nd, 5th and 7th element
select <- c(2, 5, 7)
vect1[select]
```

```
[1]  4  3 10
```

```
# extract elements in position 1 to 4, 6 and 9
select <- c(1:4, 6, 9)
vect1[select]
```

```
[1] 5 4 2 8 1 6
```

Try it in your browser

Edit the code below and press **Run**. It executes entirely in your browser via WebR — no R installation needed. The runtime downloads once, on this page only.

```
{webr-r}
vect <- c(12, 7, 25, 3, 18, 9)

# single element, a range, and specific positions
vect[2]
vect[2:4]
vect[c(1, 3, 6)]

# negative indices drop elements
vect[-c(1, 2)]

# logical subsetting
vect[vect > 10]
```

Subset Named Vectors

Vectors can be subset using the name of the elements. **When using name of elements for subsetting, ensure that the names are enclosed in single or double quotations**, else R will return an error. Let us look at a few examples:

```
vect1 <- c(score1 = 8, score2 = 6, score3 = 9)
vect1
```

```
score1 score2 score3
     8      6      9
```

```
# extract score2
vect1['score2']
```

```
score2
     6
```

```
# extract score1 and score3
vect1[c('score1', 'score3')]
```

```
score1 score3
     8      9
```

Subset using logical values

Logical values can be used to subset vectors. They are not very flexible but can be used for simple indexing. In all of the below examples, the logical vectors are recycled to match the length of the vector from which we subset data:

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  6  2  4  3 10  5  9  7  8  1
```

```
# returns all values
vect1[TRUE]
```

```
 [1]  6  2  4  3 10  5  9  7  8  1
```

```
# empty vector
vect1[FALSE]
```

```
integer(0)
```

```
# values in odd positions
vect1[c(TRUE, FALSE)]
```

```
[1]  6  4 10  9  8
```

```
# values in even positions
vect1[c(FALSE, TRUE)]
```

```
[1] 2 3 5 7 1
```

Subset using logical expressions

Logical expressions can be used to extract elements that meet specific criteria. This method is most flexible and useful as we can combine multiple conditions using relational and logical operators. Before we use logical expressions, let us spend some time understanding comparison and logical operators as we will be using them extensively hereafter.

Comparison Operators

When you create an expression using a comparison operator, the output is always a logical value i.e. `TRUE` or `FALSE`. Let us see how we can use comparison operators to subset data:

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1] 10  8  5  1  7  4  6  2  9  3
```

```
# return elements greater than 5
vect1 > 5
```

```
 [1]  TRUE  TRUE FALSE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
```

```
vect1[vect1 > 5]
```

```
[1] 10  8  7  6  9
```

```
# return elements greater than or equal to 5
vect1 >= 5
```

```
 [1]  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE
```

```
vect1[vect1 >= 5]
```

```
[1] 10  8  5  7  6  9
```

```
# return elements lesser than 5
vect1 < 5
```

```
 [1] FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE
```

```
vect1[vect1 < 5]
```

```
[1] 1 4 2 3
```

```
# return elements lesser than or equal to 5
vect1 <= 5
```

```
 [1] FALSE FALSE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE
```

```
vect1[vect1 <= 5]
```

```
[1] 5 1 4 2 3
```

```
# return elements equal to 5
vect1 == 5
```

```
 [1] FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
vect1[vect1 == 5]
```

```
[1] 5
```

```
# return elements not equal to 5
vect1 != 5
```

```
 [1]  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
```

```
vect1[vect1 != 5]
```

```
[1] 10  8  1  7  4  6  2  9  3
```

Logical Operators

Let us combine comparison and logical operators to create expressions and use them to subset vectors:

```
# random sample of 10 values
vect1 <- sample(10)
vect1
```

```
 [1]  4  9  3  8  5  6 10  2  1  7
```

```
# return all elements less than 8 or divisible by 3
vect1[(vect1 < 8 | (vect1 %% 3 == 0))]
```

```
[1] 4 9 3 5 6 2 1 7
```

```
# return all elements less than 7 or divisible by 2
vect1[(vect1 < 7 | (vect1 %% 2 == 0))]
```

```
[1]  4  3  8  5  6 10  2  1
```

Practice Check

Knowledge Check: *Index positions start at 1* What does `vect1[3]` return, and what does `vect1[0]` return?

Click to reveal answer **Answer:** `vect1[3]` returns the **third** element. `vect1[0]` returns an empty vector.

Explanation: R counts positions starting at 1, so the first element is `vect1[1]`. Position 0 does not exist, so you get nothing back rather than the first element the way a 0-based language would give you. **Knowledge Check:** *Out of range indices* `vect1` has 10 elements. What does `vect1[11]` return?

Click to reveal answer **Answer:** `NA`.

Explanation: Index 0 returns an empty vector, but an index that is past the end of the vector returns `NA` instead. An empty result means "you asked for a position that cannot exist"; `NA` means "that element is missing". **Knowledge Check:** *Negative indices delete* What does `vect1[-3]` do — return the third element from the end, or something else?

Click to reveal answer **Answer:** It **drops** the third element and returns everything else.

Explanation: Unlike Python and most other languages, R does not count backwards from the end for you. A negative index removes that position, so `vect1[-3]` is "all elements except the third". Use `tail(vect1, 3)` if you actually want the last three. **Knowledge Check:** *Coercion to the most flexible type* What is the data type of `c(1, 1L, 'one', TRUE)`?

Click to reveal answer **Answer:** `character`, giving `c("1", "1", "one", "TRUE")`.

Explanation: Coercion runs `logical` → `integer` → `double/numeric` → `character`, always toward the most flexible type. The single `'one'` forces everything to `character`. **Knowledge Check:** *Coercion without a character element* What is the data type of `c(1, 1L, TRUE)`?

Click to reveal answer **Answer:** `numeric` (double), giving `c(1, 1, 1)`.

Explanation: With no `character` present, the most flexible type available is `numeric`, so the integer and the logical are promoted to double. You only fall back to `integer` when the data contains neither `character` nor `numeric`, as in `c(1L, TRUE)`. **Knowledge Check:** *Missing values in a computation* What does `mean(c(1, 3, NA, 5, 2))` return?

Click to reveal answer **Answer:** `NA`. Use `mean(c(1, 3, NA, 5, 2), na.rm = TRUE)` to get `2.75`.

Explanation: `NA` means "unknown", not "zero", so one unknown value makes the whole mean unknown. `na.rm = TRUE` drops the missing values first, leaving 1, 3, 5, 2 — whose mean is 11 / 4 = 2.75. Decide deliberately whether dropping missing values is appropriate for your data.

Dataframes in R

Learning Objectives: After this chapter you will be able to:

- Create a data frame from equal-length vectors.
- Inspect one with `str()`, `dim()`, `names()`, and `nrow()`/`ncol()`.
- Select columns with `[`, `[[`, and `$`.
- Select rows by position and by condition.
- Subset rows and columns independently.
- Explain why `stringsAsFactors` defaults to `FALSE` in R 4.0.0 and later.

Introduction

Data frames are lists of equal-length vectors, so they can hold columns of different types.

Create dataframes

Use `data.frame` to create dataframes. Below is the function syntax:

```
args(data.frame)
```

```
function (..., row.names = NULL, check.rows = FALSE, check.names = TRUE,
    fix.empty.names = TRUE, stringsAsFactors = FALSE)
NULL
```

Data frames are basically lists with elements of equal length and as such, they are heterogeneous. Let us create a dataframe:

```
name <- c('John', 'Jack', 'Jill')
age <- c(29, 25, 27)
graduate <- c(TRUE, TRUE, FALSE)
students <- data.frame(name, age, graduate)
students
```

```
  name age graduate
1 John  29     TRUE
2 Jack  25     TRUE
3 Jill  27    FALSE
```

Basic Information

```
class(students)
## [1] "data.frame"
names(students)
## [1] "name"     "age"      "graduate"
colnames(students)
## [1] "name"     "age"      "graduate"
str(students)
## 'data.frame':    3 obs. of  3 variables:
##  $ name    : chr  "John" "Jack" "Jill"
##  $ age     : num  29 25 27
##  $ graduate: logi  TRUE TRUE FALSE
dim(students)
## [1] 3 3
nrow(students)
## [1] 3
ncol(students)
## [1] 3
```

Select Columns

Single Column

- []
- [[]]
- $

```
# using [
students[1]
##   name
## 1 John
## 2 Jack
## 3 Jill

# using [[
students[[1]]
## [1] "John" "Jack" "Jill"

# using $
students$name
## [1] "John" "Jack" "Jill"
```

Multiple Columns

```
students[, 1:3]
##   name age graduate
```

```
## 1 John  29     TRUE
## 2 Jack  25     TRUE
## 3 Jill  27    FALSE

students[, c(1, 3)]
##   name graduate
## 1 John     TRUE
## 2 Jack     TRUE
## 3 Jill    FALSE
```

Select Rows

```
# single row
students[1, ]
##   name age graduate
## 1 John  29     TRUE

# multiple row
students[c(1, 3), ]
##   name age graduate
## 1 John  29     TRUE
## 3 Jill  27    FALSE
```

Try it in your browser

Edit the code below and press **Run**. It executes entirely in your browser via WebR — no R installation needed. The runtime downloads once, on this page only.

```
{webr-r}
students <- data.frame(
  name = c("Asha", "Ravi", "Meera", "Arjun"),
  age = c(21, 22, 20, 23),
  graduate = c(TRUE, TRUE, FALSE, TRUE)
)

# rows and columns independently
students[2, ]
students[, "age"]

# filter by condition
students[students$age > 21, ]

# how many rows and columns
dim(students)
```

Starting with R 4.0.0 (April 2020), the default value of `stringsAsFactors` in `data.frame()` is `FALSE`, meaning character columns remain as character type. In older versions of R (< 4.0.0), the default was `TRUE`, which automatically converted character vectors to factors.

Common Mistake: Assuming `data.frame()` still converts strings to factors. That was the default before R 4.0.0 and was a frequent source of confusion. On R >= 4.0.0 the default is `FALSE`.

```
students <- data.frame(name, age, graduate, stringsAsFactors = FALSE)
```

Practice Check

Knowledge Check: *One column, three operators* What do `students[1]`, `students[[1]]`, and `students$name` each return?

Click to reveal answer **Answer:** A one-column **data frame**; a **vector**; and the same **vector**.

Explanation: `[` always returns a data frame, even for a single column, so downstream code that expects a vector breaks. `[[` and `$` both extract the column itself as a vector. Reach for `[[` or `$` whenever you want the values. **Knowledge Check:** *Rows and columns are indexed separately* What does `students[1, ]` return, and how does that differ from `students[, 1]`?

Click to reveal answer **Answer:** `students[1, ]` returns the **first row**; `students[, 1]` returns the **first column**. The comma is what separates the two.

Explanation: Inside `[` the first position selects rows and the second selects columns. Leaving one side empty means "all of them". So `students[1, ]` is row 1, `students[, 1]` is column 1, and `students[1, 1]` is the single cell where they cross. **Knowledge Check:** *Filtering rows by a condition* How do you return only the students older than 26?

Click to reveal answer **Answer:** `students[students$age > 26, ]`, which gives John (29) and Jill (27).

Explanation: The condition builds a logical vector with one value per row. Using it as the row index keeps the rows where the condition is `TRUE` and drops the rest. Because the condition has the same length as the data frame, the column is left out of the result. **Knowledge Check:** *How many rows and columns?* The data frame has three columns and three rows. Which two functions report those counts directly?

Click to reveal answer **Answer:** `nrow(students)` gives `3` and `ncol(students)` gives `3`. Both are also available together as `dim(students)`, which returns `c(3, 3)`.

Explanation: `dim()` returns rows first, then columns, so `dim(students)[1]` is the row count. `str()` gives you a structural overview while `nrow()` and `ncol()` answer the two specific questions directly. **Knowledge Check:** *`stringsAsFactors` changed in R 4.0.0* In R 4.0.0 and later, what does `data.frame(name, age, graduate)` do with the character column `name`?

Click to reveal answer **Answer:** It keeps it as `character`. Before R 4.0.0 the default was `TRUE`, which silently converted character columns to factors.

Explanation: The old default caused endless confusion, because strings became categories and levels were alphabetised behind your back. Passing `stringsAsFactors = FALSE` is now redundant on R >= 4.0.0, but it keeps code readable and portable to older R.

Factors in R

Learning Objectives: After this chapter you will be able to:

- Create a factor from a character vector.
- Inspect and count levels with `levels()` and `nlevels()`.
- Impose a natural order with `levels=` together with `ordered = TRUE`.
- Attach shorter display labels with `labels=`.
- Test for ordering with `is.ordered()`.
- Reach for `forcats` when you need to recode, reorder, or lump levels.

Introduction

Categorical or qualitative data in R is treated as data type `factor`.

Create Factors

```
args(factor)
```

```
function (x = character(), levels, labels = levels, exclude = NA,
    ordered = is.ordered(x), nmax = NA)
NULL
```

```
devices <- factor(c('Mobile', 'Tablet', 'Desktop'))
devices
```

```
[1] Mobile  Tablet  Desktop
Levels: Desktop Mobile Tablet
```

```
# number of levels
nlevels(devices)
```

```
[1] 3
```

```
# levels
levels(devices)
```

```
[1] "Desktop" "Mobile"  "Tablet"
```

Order Levels

Use the `levels` argument together with `ordered = TRUE` when the categories have a natural order:

```
size <- factor(c('Medium', 'Low', 'High', 'Low', 'Medium'),
  levels = c('Low', 'Medium', 'High'), ordered = TRUE)
size
```

```
[1] Medium Low    High   Low    Medium
Levels: Low < Medium < High
```

```
levels(size)
```

```
[1] "Low"    "Medium" "High"
```

Specify Labels

Use the `labels` argument to attach shorter or prettier display labels while keeping the underlying level order:

```
size_labeled <- factor(c('Low', 'Medium', 'High'),
  levels = c('Low', 'Medium', 'High'), labels = c('L', 'M', 'H'))
size_labeled
```

```
[1] L M H
Levels: L M H
```

Check Levels

```
# existing levels
levels(size)
```

```
[1] "Low"    "Medium" "High"
```

```
# number of levels
nlevels(size)
```

```
[1] 3
```

```
# is this an ordered factor?
is.ordered(size)
```

```
[1] TRUE
```

For modern factor manipulation (reordering, recoding, lumping rare levels), see the forcats package.

Lists in R

Learning Objectives: After this chapter you will be able to:

- Create a list that holds elements of different types.
- Name list elements with `names()`.
- Access elements with `[[`, `[`, and `$`.
- Explain why `[[` returns the element while `[` returns a list.
- Convert a vector or matrix into a list with `as.list()`.
- Convert a list back into a vector with `unlist()`.

Introduction

Lists are very useful as they are heterogeneous i.e. they can contain different data types. If you remember, vectors and matrices are homogeneous i.e. they can contain only one type of data. If you include different data types, they will all be coerced to the same type. With lists, it is different. In this chapter, we will delve deeper into lists.

Creating Lists

To create a list, we use the `list()` function. Let us create a simple list to demonstrate how they can contain different data types.

```
# numeric vector
vect1 <- 1:10

# character vector
vect2 <- c('Jack', 'John', 'Jill')

# logical vector
vect3 <- c(TRUE, FALSE)

# matrix
mat <- matrix(data = 1:9, nrow = 3)

# list
l <- list(vect1, vect2, vect3, mat)
l
```

```
[[1]]
 [1]  1  2  3  4  5  6  7  8  9 10

[[2]]
[1] "Jack" "John" "Jill"

[[3]]
[1]  TRUE FALSE

[[4]]
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
```

If you observe the output, all the elements of the list retain their data types. Now let us learn how to access the elements of the list.

Access List Elements

You can access the elements of a list using the following operators:

- `[[`
- `[`
- `$`

Let us try them one by one.

```
# using [[
l[[1]]
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
# using [
l[1]
```

```
[[1]]
 [1]  1  2  3  4  5  6  7  8  9 10
```

`[[` extracts the element itself (which may be a vector, matrix, data frame, or nested list), while `[` returns a new list containing the selected element(s). The `$` operator can be used only when we have named elements in the list. Let us add names to the elements. Use the `names()` function to add names to the list.

```
# named elements
names(l) <- c('vect1', 'vect2', 'vect3', 'mat')
l
```

```
$vect1
 [1]  1  2  3  4  5  6  7  8  9 10

$vect2
[1] "Jack" "John" "Jill"

$vect3
[1]  TRUE FALSE

$mat
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
```

```
# use $
l$vect1
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
# use [[
l[['vect1']]
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
# use [
l['vect1']
```

```
$vect1
 [1]  1  2  3  4  5  6  7  8  9 10
```

Coerce other objects

You can coerce other objects to list using `as.list()`.

```
# vector
vect1 <- 1:10
as.list(vect1)
```

```
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] 3

[[4]]
[1] 4

[[5]]
[1] 5

[[6]]
[1] 6

[[7]]
[1] 7

[[8]]
[1] 8

[[9]]
[1] 9

[[10]]
[1] 10
```

```
# matrix
mat <- matrix(data = 1:9, nrow = 3)
as.list(mat)
```

```
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] 3

[[4]]
[1] 4

[[5]]
[1] 5

[[6]]
```

```
[1] 6

[[7]]
[1] 7

[[8]]
[1] 8

[[9]]
[1] 9
```

Coerce list to other objects

Use `unlist()` to coerce a list to a vector.

```
# unlist
unlist(l)
```

```
 vect11  vect12  vect13  vect14  vect15  vect16  vect17  vect18  vect19 vect110
    "1"     "2"     "3"     "4"     "5"     "6"     "7"     "8"     "9"    "10"
 vect21  vect22  vect23  vect31  vect32    mat1    mat2    mat3    mat4    mat5
 "Jack"  "John"  "Jill"  "TRUE" "FALSE"     "1"     "2"     "3"     "4"     "5"
   mat6    mat7    mat8    mat9
    "6"     "7"     "8"     "9"
```

```
# remove names
unlist(l, use.names = FALSE)
```

```
 [1] "1"     "2"     "3"     "4"     "5"     "6"     "7"     "8"     "9"
[10] "10"    "Jack"  "John"  "Jill"  "TRUE"  "FALSE" "1"     "2"     "3"
[19] "4"     "5"     "6"     "7"     "8"     "9"
```

Practice Check

Knowledge Check: *Why use a list at all?* `vect1 <- 1:10`, `vect2 <- c('Jack', 'John', 'Jill')`, and `vect3 <- c(TRUE, FALSE)`. Why not just `c(vect1, vect2, vect3)`?

Click to reveal answer **Answer:** That would coerce everything to `character`, because `character` is the most flexible type. A list keeps each element's own type.

Explanation: `c(vect1, vect2, vect3)` gives you `c("1", "2", ..., "10", "Jack", ...)`. Vectors and matrices are homogeneous, so mixing types destroys information. `list(vect1, vect2, vect3)` stores the three vectors side by side with their types intact. **Knowledge Check:** *`[[` versus `[`* Given `l <- list(1:10, c('Jack', 'John'))`, what is the difference between `l[[1]]` and `l[1]`?

Click to reveal answer **Answer:** `l[[1]]` is the vector `1:10` itself. `l[1]` is a list containing one element, which is that same vector.

Explanation: `[[` extracts the element, so you get the object and can operate on it directly. `[` builds a new list holding the selection, which is what you want when you want to delete or keep several elements. Square brackets are for selecting; double square brackets are for extracting. **Knowledge**

Check: *`$` needs names* Why does `l$vect1` fail before you run `names(l) <- c('vect1', 'vect2', 'vect3', 'mat')`?

Click to reveal answer **Answer:** The elements have no names, so there is nothing for `$` to match.

Explanation: `$` looks up elements **by name**, not by position. Until the list has names, use `l[[1]]` or `l[1]`. Once named, `l$vect1` and `l[['vect1']]` both return the vector. **Knowledge Check:** *Coercing to a list* How do you turn the vector `vect1 <- 1:10` into a list, one element per value?

Click to reveal answer **Answer:** `as.list(vect1)`, which gives a list of ten single numbers.

Explanation: `as.list()` splits the vector into its individual elements. The same call works on a matrix, in which case it splits the matrix into columns. The reverse direction is `unlist()`. **Knowledge Check:** *Flattening a list* What does `unlist(l, use.names = FALSE)` do differently from `unlist(l)`?

Click to reveal answer **Answer:** It returns the same values but without the names.

Explanation: `unlist()` concatenates every element into one vector and keeps the names it finds, so you get a named vector. Setting `use.names = FALSE` drops the names, which is usually what you want for further calculation.

Install & Update R Packages

Learning Objectives: After this chapter you will be able to:

- Install packages from CRAN, GitHub, Bitbucket, and Bioconductor.
- Install a specific archived version of a package.
- List, update, and remove installed packages.
- Find library paths with `.libPaths()` instead of hardcoding a path.

Introduction

In this chapter, we will learn about R packages. Packages are fundamental to R. There are more than 22,000 packages available on CRAN or the Comprehensive R Archive Network (as of September 2026; verify the current count at cran.r-project.org/web/packages).

Packages are available for different topics. You should always look for a package before writing code from scratch. In case you have written your own codes for a new analysis or topic, do share it with the R community by converting the code into a package. You can learn more about building R packages from R Packages, a book written by Hadley Wickham.

Install Packages

CRAN

Packages from CRAN can be installed using `install.packages()`. The name of the package must be enclosed in single or double quotes.

```
install.packages('ggplot2')
```

GitHub

Some R packages are made available on GitHub before releasing them on CRAN. Such packages can be installed using `install_github()` from devtools or remotes package. You need to specify the name of the repository and the package. For example, to download ggplot2 or dplyr, below is the code:

```
devtools::install_github("tidyverse/ggplot2")
remotes::install_github("tidyverse/dplyr")
```

BitBucket

Bitbucket is similar to GitHub. You can install packages from Bitbucket using `install_bitbucket()` from devtools or remotes package.

```
devtools::install_bitbucket("dannavarro/lsr-package")
remotes::install_bitbucket("dannavarro/lsr-package")
```

Bioconductor

Bioconductor provides tools for analysis and comprehension of high throughput genomic data. Install the `BiocManager` package and use `BiocManager::install()`:

```
# install BiocManager if needed
install.packages("BiocManager")

# install a Bioconductor package
BiocManager::install("GenomicFeatures")
```

rForge

Many R packages are hosted at R-Forge, a platform for development of R packages. (Note: R-Forge is largely inactive today; prefer CRAN or GitHub for new work.)

```
install.packages('quantstrat', repos = 'https://r-forge.r-project.org/')
```

> For faster installs with dependency resolution, you can also try `pak::pkg_install()` from the pak package as a modern alternative.

Install Different Versions

Now that we have learnt how to install packages, let us look at installing different versions of the same package.

```
remotes::install_version('dplyr', version = 0.5.0)
```

If you want to install the latest release from GitHub, append `@*release` to the repository name. For example, to install the latest release of dplyr:

```
remotes::install_github('tidyverse/dplyr@*release')
```

Installed Packages

- `installed.packages()`: view currently installed packages
- `library('package_name')`: load packages
- `autoload('function_name', 'package_name')`: load functions and data from packages only when called in the script
- `available.packages()`: packages available for installation
- `old.packages()`: packages which have new versions available
- `new.packages()`: packages already not installed
- `update.packages()`: update packages which have new versions
- `remove.packages('package_name')`: remove installed packages

Library Paths

Library is a directory that contains all installed packages. Usually there will be more than one R library in your system. You can find the location of the libraries using `.libPaths()`.

```
.libPaths()
```

```
[1] "/home/runner/.cache/R/renv/library/intro-r-eca02dad/linux-ubuntu-noble/
R-4.5/x86_64-pc-linux-gnu"
[2] "/home/runner/.cache/R/renv/sandbox/linux-ubuntu-noble/R-4.5/x86_64-pc-linux-
gnu/46e831e3"
```

You can use `lib` when you want to install, load, update and remove packages from a particular library.

Install

```
install.packages('stringr', lib = .libPaths()[1])
```

Load

```
library(lubridate, lib = .libPaths()[1])
```

Update Packages

```
update.packages(lib = .libPaths()[1])
```

Remove Packages

```
remove.packages('stringr', lib = .libPaths()[1])
```

Explore Rsquared Academy's own R packages and guides at pkgs.rsquaredacademy.com.

RStudio Projects & Data Import

Learning Objectives: After this chapter you will be able to:

- Work inside an RStudio Project (`.Rproj`) instead of `setwd()`.
- Build portable paths with `here::here()`.

- Read CSV files with `readr::read_csv()` and Excel files with `readxl::read_excel()`.

```
library(readr)
library(readxl)
library(here)
```

Why avoid `setwd()`

`setwd("C:/Users/you/project")` hard-codes a path that exists only on your machine. Anyone else who opens the script — or you on a different computer — gets an error. An RStudio Project (`.Rproj`) fixes this: opening the `.Rproj` file sets the working directory to the project folder automatically, so relative paths just work.

Workflow: File → New Project → New Directory (or Existing Directory) → give it a name. From then on, open the `.Rproj` file, not individual scripts.

Portable paths with `here`

The `here` package builds paths from the project root, wherever the project lives on disk:

```
here("data", "students.csv")
```

```
[1] "/home/runner/work/intro-r/intro-r/data/students.csv"
```

> **Common Mistake:** Paths with `../` or `C:/...` break when the project moves. If a path starts with a drive letter or `../`, rewrite it with `here()`.

Reading CSV files

Practice data lives in the `data/` directory (CC0 / public domain, created for this book). Read it with `read_csv()`:

```
students <- read_csv(here("data", "students.csv"))
students
```

```
# A tibble: 5 x 3
  name    age score
  <chr> <dbl> <dbl>
1 Asha     21    88
2 Ravi     22    92
3 Meera    20    79
4 Arjun    23    95
5 Divya    21    84
```

Reading Excel files

`read_excel()` works the same way for `.xlsx` files — sheet 1 by default:

```
students_xl <- read_excel(here("data", "students.xlsx"))
students_xl
```

```
# A tibble: 5 x 3
  name    age score
```

```
  <chr> <dbl> <dbl>
1 Asha     21    88
2 Ravi     22    92
3 Meera    20    79
4 Arjun    23    95
5 Divya    21    84
```

Try it in your browser

Edit the code below and press **Run**. It executes entirely in your browser via WebR — no R installation needed. The runtime downloads once, on this page only.

```
{webr-r}
library(readr)

# the browser filesystem starts empty, so fetch the sample first
if (!file.exists("students.csv")) {
  download.file(
    "https://intro-r.rsquaredacademy.com/data-webr/students.csv",
    "students.csv"
  )
}

students <- read_csv("students.csv", show_col_types = FALSE)
students

# column types are parsed, not guessed at read time
sapply(students, class)
```

Practice Check

Knowledge Check: *Project-relative paths* Why does `read_csv(here("data", "students.csv"))` keep working when you move the whole project folder to another computer, but `read_csv("C:/Users/you/project/data/students.csv")` does not?

Click to reveal answer **Answer:** `here()` resolves the path relative to the project root at run time, so it adapts to the new location. The `C:/...` path is absolute and points to a folder that does not exist on the other computer.

Explanation: Projects + `here()` keep every path portable (`project root` → `data` → `file`); absolute paths and `setwd()` tie the script to one machine. **Knowledge Check:** *Why `.Rproj` beats `setwd()`* What breaks when you share a script containing `setwd("C:/Users/you/project")` with a colleague?

Click to reveal answer **Answer:** The script errors because that folder does not exist on your colleague's machine.

Explanation: `setwd()` with an absolute path ties the script to one computer; opening a shared `.Rproj` file sets everyone's working directory to the project folder automatically. **Knowledge Check:** *`here()` resolution* Your project lives at `D:/work/survey` and contains `data/resp.csv`. What does `here("data", "resp.csv")` return?

Click to reveal answer **Answer:** `D:/work/survey/data/resp.csv` (i.e. `project root` → `data` → `resp.csv`).

Explanation: `here()` anchors paths at the project root (where the `.Rproj` file lives), joining each additional component with the correct separator for your OS. **Knowledge Check:** *`read_csv()` vs `read.csv()`* Name one practical advantage of `readr::read_csv()` over base R's `read.csv()` for a beginner.

Click to reveal answer **Answer:** Any one of: it never converts strings to factors, it prints a compact tibble preview, or it reports column types as it reads.

Explanation: `read_csv()` uses modern defaults (no `stringsAsFactors` surprises) and tells you the parsed type of each column, so mismatches are visible immediately. **Knowledge Check:** *Excel sheets* `read_excel(here("data", "students.xlsx"))` reads which sheet by default, and how would you read the second sheet instead?

Click to reveal answer **Answer:** The first sheet by default; pass `sheet = 2` (or the sheet name) to read the second sheet.

Explanation: `read_excel()` defaults to sheet 1; the `sheet` argument selects another sheet by position or name.

Your First Visualization

Learning Objectives: After this chapter you will be able to:

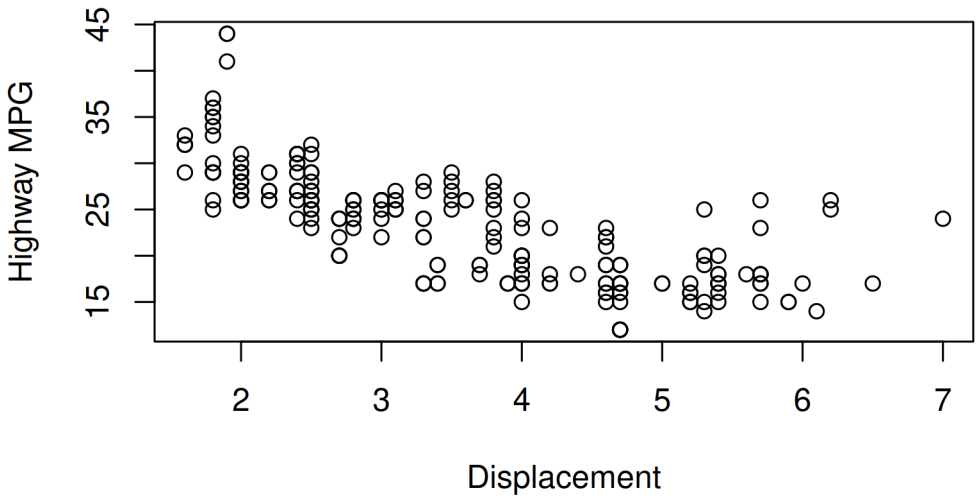
- Load `ggplot2` and its bundled `mpg` dataset.
- Draw the same scatterplot in base R and in ggplot2.
- Describe a plot as data + aesthetics + geometry.

```
library(ggplot2)
# ggplot2 bundles the `mpg` dataset (MIT license, same as ggplot2 itself)
data(mpg, package = "ggplot2")
```

Common Mistake: Base R has no `mpg` dataset (it has `mtcars`). `plot(mpg$displ, mpg$hwy)` without `library(ggplot2)` fails with `object 'mpg' not found`. Always load `ggplot2` first.

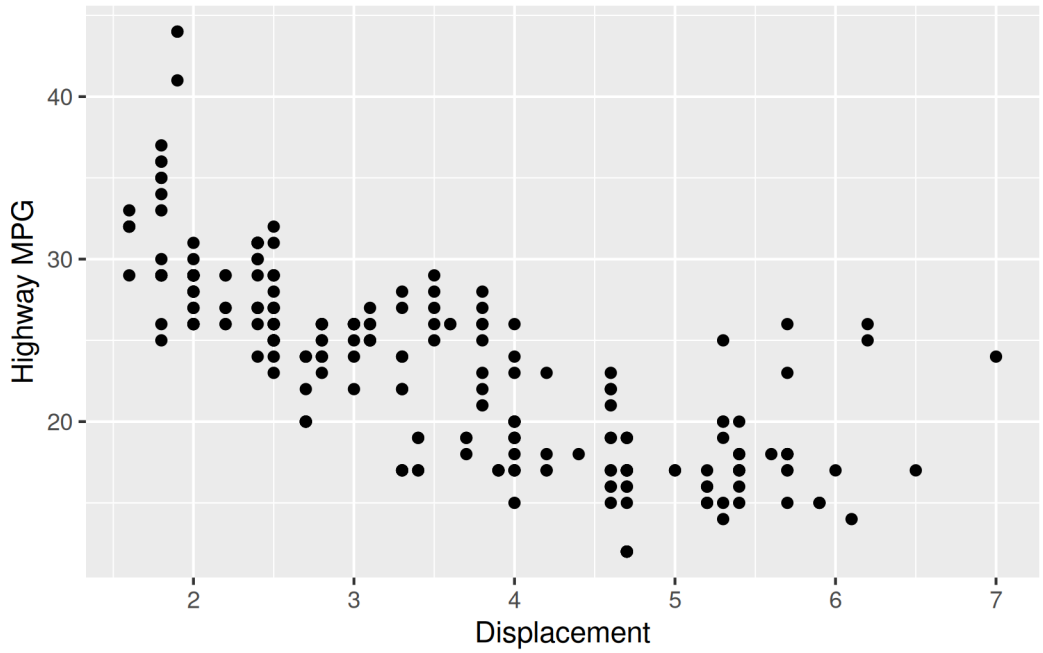
Base R graphics

```
plot(mpg$displ, mpg$hwy, xlab = "Displacement", ylab = "Highway MPG")
```



ggplot2 graphics

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  labs(x = "Displacement", y = "Highway MPG")
```



Mental model: every ggplot is *data* (`mpg`) + *aesthetics* (`aes(x, y)`) + *geometry* (`geom_point()`). Change the geometry, keep the mapping.

Try it in your browser

Edit the code below and press **Run**. It executes entirely in your browser via WebR — no R installation needed. The runtime downloads once, on this page only.

```
{webr-r}
library(ggplot2)

p <- ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(color = "steelblue") +
  labs(x = "Displacement", y = "Highway MPG")
p

# same data and mapping, different geometry
p + geom_smooth(method = "lm", se = FALSE)
```

Practice Check

Knowledge Check: *Data + aesthetics + geometry* In `ggplot(mpg, aes(x = displ, y = hwy)) + geom_point()`, which part is the data, which is the aesthetics mapping, and which is the geometry?

Click to reveal answer **Answer:** Data: `mpg`. Aesthetics: `aes(x = displ, y = hwy)`. Geometry: `geom_point()`.

Explanation: `ggplot()` takes the data and the variable mapping; each `geom_*()` layer decides how those mapped variables are drawn. **Knowledge Check:** *Why `library(ggplot2)` comes first* What error do you get from `plot(mpg$displ, mpg$hwy)` in a fresh R session, and why?

Click to reveal answer **Answer:** `object 'mpg' not found` — base R has no `mpg` dataset (it has `mtcars`); `mpg` only exists after `library(ggplot2)` (or `data(mpg, package = "ggplot2")`).

Explanation: Datasets live inside packages. Until the package is loaded, its data is invisible to your session. **Knowledge Check:** *Changing the geometry* How would you turn the scatterplot into a smoothed trend line plot without changing the data or the variable mapping?

Click to reveal answer **Answer:** Replace `geom_point()` with `geom_smooth()` (or add it as a second layer).

Explanation: Data + aesthetics stay fixed while geometries are interchangeable layers — that is the point of the grammar of graphics. **Knowledge Check:** *Aesthetics mapping* In `aes(x = displ, y = hwy)`, what do `x` and `y` refer to — plot labels or dataset columns?

Click to reveal answer **Answer:** Dataset columns: `mpg$displ` mapped to horizontal position, `mpg$hwy` to vertical position.

Explanation: `aes()` maps columns of the data to visual properties. Labels are set separately with `labs()`. **Knowledge Check:** *Base R vs ggplot2* The base R call repeats `mpg$` twice; the ggplot2 call mentions `mpg` once. Why is that difference useful as plots get more complex?

Click to reveal answer **Answer:** Declaring the data once in `ggplot(mpg, ...)` means every layer inherits it — you never retype the data source per variable, so longer plots stay readable and less error-prone.

Explanation: One data declaration shared across layers beats repeating `data$column` for every variable, especially with facets and multiple geoms.

Your First Data Wrangling

Learning Objectives: After this chapter you will be able to:

- Use the five core `dplyr` verbs: `filter()`, `select()`, `mutate()`, `arrange()`, `summarise()`.
- Chain steps with the native base pipe `|>`.
- Know where to go next after this book.

```
library(dplyr)
# ggplot2 bundles the `mpg` dataset (MIT license, same as ggplot2 itself)
data(mpg, package = "ggplot2")
```

We use the native base pipe `|>` (standard since R 4.1.0; zero external dependencies). We do **not** use `%>%`.

The five verbs

```
mpg |>
  filter(manufacturer == "audi") |>
  select(manufacturer, model, displ, hwy) |>
  mutate(km_per_litre = hwy * 0.425144) |>
  arrange(desc(hwy))
```

```
# A tibble: 18 x 5
   manufacturer model      displ   hwy km_per_litre
   <chr>        <chr>      <dbl> <int>        <dbl>
 1 audi         a4           2      31        13.2
 2 audi         a4           2      30        12.8
 3 audi         a4           1.8    29        12.3
 4 audi         a4           1.8    29        12.3
 5 audi         a4 quattro   2      28        11.9
 6 audi         a4           3.1    27        11.5
 7 audi         a4 quattro   2      27        11.5
 8 audi         a4           2.8    26        11.1
 9 audi         a4           2.8    26        11.1
10 audi         a4 quattro   1.8    26        11.1
11 audi         a4 quattro   1.8    25        10.6
12 audi         a4 quattro   2.8    25        10.6
13 audi         a4 quattro   2.8    25        10.6
14 audi         a4 quattro   3.1    25        10.6
15 audi         a4 quattro   3.1    25        10.6
```

```
16 audi         a6 quattro   3.1    25        10.6
17 audi         a6 quattro   2.8    24        10.2
18 audi         a6 quattro   4.2    23         9.78
```

```
mpg |>
  summarise(mean_hwy = mean(hwy), n = dplyr::n())
```

```
# A tibble: 1 x 2
  mean_hwy     n
     <dbl> <int>
1     23.4   234
```

> **Common Mistake:** `mean()` on a vector with `NA` returns `NA`. Add `na.rm = TRUE` when missing values are present.

Try it in your browser

Edit the code below and press **Run**. It executes entirely in your browser via WebR — no R installation needed. The runtime downloads once, on this page only.

```
{webr-r}
library(dplyr)

# filter, select, mutate, arrange
mpg |>
  filter(manufacturer == "audi") |>
  select(model, hwy) |>
  mutate(km_per_litre = hwy * 0.425144) |>
  arrange(desc(hwy)) |>
  head(5)

# summarise collapses to a single row
mpg |>
  summarise(mean_hwy = mean(hwy), n = n())
```

Practice Check

Knowledge Check: *Filter then select* What does `mpg |> filter(manufacturer == "audi") |> select(model, hwy)` return?

Click to reveal answer **Answer:** A data frame with two columns (`model`, `hwy`) containing only the rows where `manufacturer` is `"audi"`.

Explanation: `filter()` keeps matching rows; `select()` keeps matching columns. The `|>` pipe passes the left-hand result into the first argument of the right-hand function. **Knowledge Check:** *`mutate()` adds columns* What does `mpg |> mutate(km_per_litre = hwy * 0.425144)` return, and how many rows does the result have compared to `mpg`?

Click to reveal answer **Answer:** The full `mpg` data with one extra column (`km_per_litre`); the row count is unchanged.

Explanation: `mutate()` adds (or modifies) columns row-wise — it never adds or removes rows. Use `filter()` to change row counts. **Knowledge Check:** *`arrange()` ordering* After `mpg |> arrange(desc(hwy))`, which kind of car sits in row 1?

Click to reveal answer **Answer:** The car with the highest highway mileage (`hwy`).

Explanation: `arrange()` sorts ascending by default; wrapping in `desc()` flips it to descending, so the maximum comes first. **Knowledge Check:** *`summarise()` collapses rows* How many rows does `mpg |> summarise(mean_hwy = mean(hwy), n = dplyr::n())` return, and why?

Click to reveal answer **Answer:** Exactly one row: the mean of `hwy` across the whole dataset and the total row count.

Explanation: Without `group_by()`, `summarise()` collapses the entire data frame into a single summary row. **Knowledge Check:** *Reading a pipeline* Read this pipeline aloud in order: `mpg |> filter(manufacturer == "audi") |> select(model, hwy) |> arrange(desc(hwy))`. What question does it answer?

Click to reveal answer **Answer:** "Among Audi models, which has the best (and worst) highway mileage?" — it keeps only Audis, keeps the model and mileage columns, and sorts best-first.

Explanation: Pipelines read top-to-bottom as a sequence of verbs; narrating each step in plain words is the fastest way to check you built the right one.

Where to go next

Where to go next: - Book: R for Data Science (2e) — start at Chapter 1 - Courses: Rsquared Academy Data Wrangling series - More books: ebooks.rsquaredacademy.com